

Figure 14.3 Circuit used in Ex. 14.1.

For the capacitor to discharge to ground (from an initial voltage of 1 V) takes, roughly, $(1 \text{ V})/(71 \text{ mV}/\mu\text{s}) = 14 \mu\text{s}$. Figure 14.4 shows the simulation results. Note how the time it takes the output voltage to discharge to ground is twice as long as what we calculated or roughly $30 \mu\text{s}$. This is due to the fact that as the voltage between the drain and source of the MOSFET decreases so does the off current. The reduction in the off current causes the capacitor to discharge slower. To keep the voltage across the capacitor from discharging too much we might try holding one side of the MOSFET switch at $V_{DD}/2$. Consider the following. ■

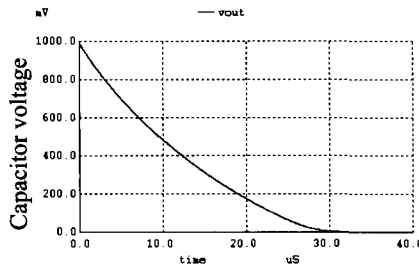


Figure 14.4 Time it takes the capacitor to discharge due to the off current of the MOSFET in Fig. 14.3.

Example 14.2

Repeat Ex. 14.1 with the input tied to $V_{DD}/2$ instead of ground.

The schematic and simulation results are seen in Fig. 14.5. The leakage current is minimized by lowering the V_{DS} of the MOSFET. This is a common technique to

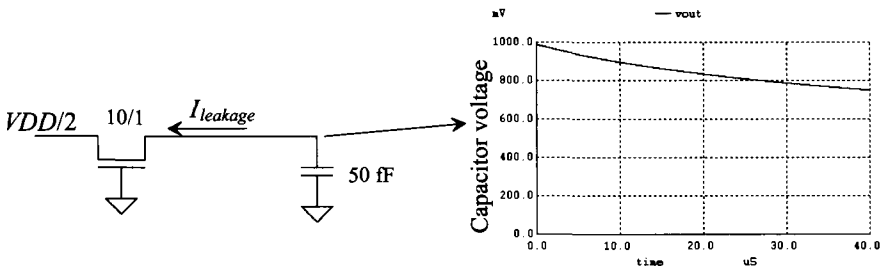


Figure 14.5 Circuit used in Ex. 14.2.

reduce leakage currents in DRAM (see Fig. 16.7 where the bitlines are equilibrated to $V_{DD}/2$). Note that when the voltage across the capacitor (the storage node) is zero and the PG is off the leakage through the MOSFET causes the storage node to charge upwards (see Problem 14.2). ■

Example 14.3

Figure 14.6 shows a dynamic level-sensitive latch. Estimate the maximum time PG can be off before data is lost on the charge storage node. Compare the estimate to SPICE. Use the 50 nm process with 20/1 PMOS devices and 10/1 NMOS devices.

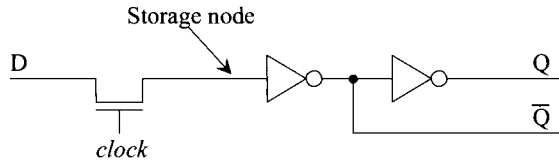


Figure 14.6 A dynamic level-sensitive latch.

The input capacitance of the inverter, using Table 10.2, is estimated as

$$C_{in} = \frac{3}{2} \cdot (1.25 + 0.625) \text{ fF} = 2.8 \text{ fF}$$

Using Eq. (14.3), we can estimate the rate the storage node decays as

$$\frac{dV}{dt} = \frac{3.55 \text{ nA}}{2.8 \text{ fF}} = 1.27 \text{ V}/\mu\text{s}$$

If we want, at most, 100 mV of droop in the storage node's voltage, then we would want to ensure that the PG is clocked, at the minimum, every 100 ns. Figure 14.7 shows the simulation results. Compared to our hand calculations, the discharge rate is considerably faster. This is due to our calculation of the input capacitance. The values used for C_{in} were the switching input capacitance (see Figs. 10.2, 10.7, and 10.8), not the static input capacitance (see Fig. 6.4). ■

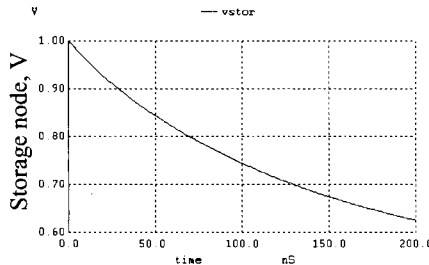


Figure 14.7 How the voltage on the storage node varies in the dynamic latch circuit seen in Fig. 14.6.

14.1.2 Simulating Dynamic Circuits

Because of the extremely small leakage currents involved, simulating dynamic circuits can be challenging. To begin, when SPICE simulates any circuit, it puts a resistor with a conductance value given by the parameter GMIN across every pn junction and MOSFET drain-to-source. The default value of GMIN is 10^{-12} mhos or a 1 T Ω resistor. A charge storage node at a potential of 1 V has a leakage current, due to GMIN, of 1 pA. The value of GMIN can be set, and increased, using the .OPTIONS command, at the cost of a longer or more difficult convergence time, to a smaller value, say 10^{-15} . Some SPICE simulators are also capable of adding a resistor at every node to ground called RSHUNT (to help with convergence). If RSHUNT is 10^9 and a node voltage is 1 V, then a current of 1 nA flows through the added RSHUNT resistor.

In practice, we can use the default values of SPICE when simulating dynamic circuits. The SPICE leakage paths (using the default values of GMIN and RSHUNT) have little effect on the estimates for the discharge times of storage nodes when simulating dynamic circuits. For example, the leakage current, for $VDD = 1$ V, due to the default value of GMIN is given by

$$I_{leakage} \approx 1 \text{ pA} = VDD \cdot GMIN \quad (14.5)$$

and

$$\frac{dV}{dt} = \frac{1 \text{ pA}}{C_{node}} = \frac{VDD \cdot GMIN}{C_{node}} \quad (14.6)$$

If $C_{node} = 5$ fF, then $dV/dt = 200 = 200 \text{ } \mu\text{V}/\mu\text{s}$. It takes approximately 5 ms for the voltage on the charge storage node to drop to ground. In all practical simulations, the leakage through a modern nanometer MOSFET is much more significant than the leakage through GMIN.

14.1.3 Nonoverlapping Clock Generation

Consider the string of PGs/inverters shown in Fig. 14.8. This circuit is called a dynamic shift register. When ϕ_1 goes high, the first and third stages of the register are enabled. Data is passed from the input to point A0 and from point A1 to A2. If ϕ_2 is low while ϕ_1 is high, the data cannot pass from A0 to A1 and from A2 to A3. If ϕ_1 goes low and ϕ_2 goes high, data is passed from A0 to A1 and from A2 to A3. If both ϕ_1 and ϕ_2 are high at the same time, the input of the shift register and the output are connected together, which is not desirable in a shift register application. The purpose of the inverter between PGs is to restore logic levels, since the NMOS PG passes a high with a threshold voltage drop. Two inverters would be used to eliminate the logic inversion between stages. *The clocks used in this dynamic circuit must be nonoverlapping, or logically*

$$\phi_1 \cdot \phi_2 = 0 \quad (14.7)$$

There should be a period of dead time between transitions of the clock signals, labeled Δ in Fig. 14.8. The rise and fall times of the clock signals should not occur at the same time.

Since the design and layout of the dynamic shift register is straightforward, let's concentrate on the generation of clock signals, ϕ_1 and ϕ_2 . Note that a simple logic inversion will not generate nonoverlapping clock signals. Consider the schematic of the nonoverlapping clock generator shown in Fig. 14.9. This circuit takes a clock signal and

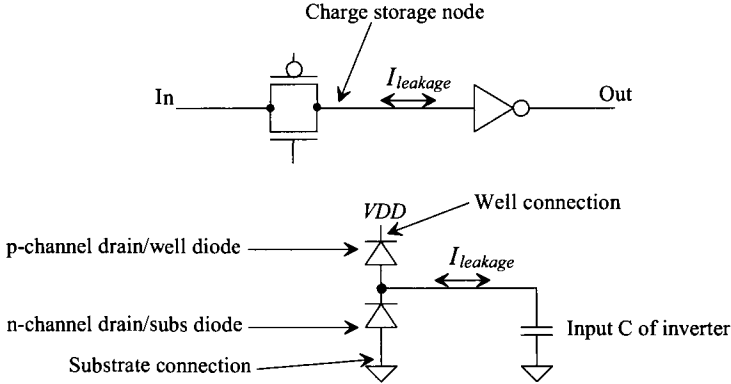


Figure 14.10 CMOS TG used in dynamic logic.

The leakage of charge off of or onto the input capacitance of the inverter in Fig. 14.10 can be attributed to the drain-well diode of the PMOS device and the drain-substrate diode of the NMOS device used in the TG. If these leakage currents were equal, then the leakage of charge off of the storage node would be zero. Notice that unlike the NMOS PG, using a TG can result in the charge storage node leaking to V_{DD} or ground, depending on the size of the drain areas and the leakage currents in each device.

14.2 Clocked CMOS Logic

In this section we provide a brief overview of dynamic, or clocked, CMOS logic design.

Clocked CMOS Latch

Figure 14.11 shows the schematic of a clocked CMOS latch. When ϕ_1 is a low, M2 and M3 are on. The master stage simply behaves like an inverter. The D input drives, through the enabled master stage, the node N1. During this time, both M6 and M7 are off. The latch's output, Q, is a charge storage node. When ϕ_1 goes high, M2 and M3 shut off and N1 is the charge storage node. M6 and M7 are on, and the Q output is actively driven either high or low.

An Important Note

Notice, in Fig. 14.11, that if the node N1 starts to move away from either V_{DD} or ground, when ϕ_1 is a high, that the slave stage can move towards its switching point. This can cause a significant current to flow from V_{DD} to ground in the slave stage. Another example of where this problem can occur is seen in Fig. 14.6. If the storage node starts to wander, the inverter's input can move towards its switching point voltage. As seen in Fig. 11.4, again, a significant current can flow.

Note that the storage node's voltage can wander because of the MOSFET's off current (Fig. 14.2), a gate tunnel current leaking into the node (Fig. 16.67), or because of capacitive coupling from a noisy node (as discussed in Ex. 3.5). The issues of gate tunneling current and the reduction in parasitic capacitances present challenges when implementing dynamic logic in nanometer CMOS technologies.

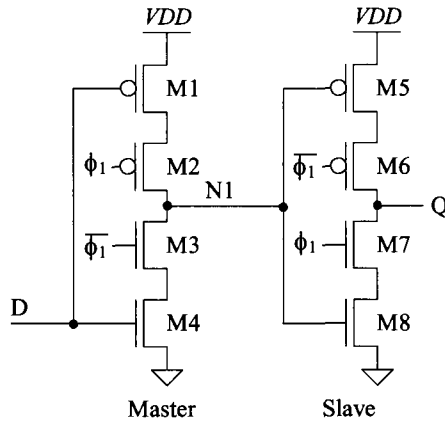


Figure 14.11 A clocked CMOS latch. The clock signals can be generated with an RS latch so that the edges occur essentially at the same moment in time.

PE Logic

This section presents precharge-evaluate logic, or PE logic. Consider the three-input NAND gate shown in Fig. 14.12. The operation of this gate relies on a single clock input. When ϕ_1 is low, the output node capacitance is charged to VDD through M5. During the evaluate phase, ϕ_1 is high, M1 is on, and if A0, A1, and A2 are high, the output is pulled low. The logic output is available only when ϕ_1 is high. The output is a logic one when ϕ_1 is low. One disadvantage of PE logic is that the gate logic output is available part of the time and not all of the time as in the static gates.

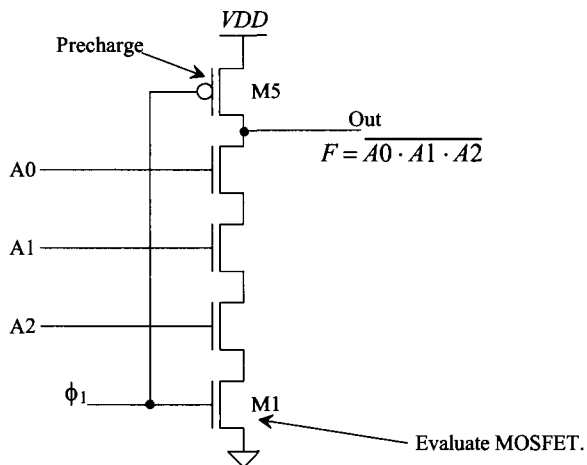


Figure 14.12 Precharge-evaluate three-input NAND gate.

Several important characteristics of the PE gate should be pointed out. The input capacitance of the PE gate is less than that of the static gate. Each input is connected to a single MOSFET where the static gate inputs are tied to two MOSFETs. Potentially the PE gate operates faster and dissipates less power.

The sizes of the MOSFETs used in a PE gate does not need adjusting for symmetrical switching point voltage. The absence of complementary devices and the fact that the output is pulled high during each half cycle makes the gate V_{sp} meaningless. However, we may need to size the devices to attain a certain speed for a given load capacitance. If the sizes of all NMOS transistors used in Fig. 14.12 are equal, then the t_{PHL} is approximately $4R_n C_{node}$ and the t_{PLH} is $R_p C_{node}$, where C_{node} is the total capacitance on the output node. This may include the interconnecting capacitance and the input capacitance of the next stage. Here we have neglected both the transmission line effects through a series connection of MOSFETs and the intrinsic switching speeds. A more complex logic function, $F = A0 + A1 \cdot A2 + A3 \cdot A4$, implemented in PE logic is shown in Fig. 14.13.

Domino Logic

Consider the cascade of PE gates shown in Fig. 14.14. During the precharge phase of the clock, the output of each PE gate is a logic high. This high-level output is connected to the input of the next PE gate. Suppose that the logic out of the first PE gate during the evaluate phase is a low. This output will turn off any MOSFETs in the second PE gate. However, during the precharge phase, those same MOSFETs in the second PE gate will be turned on. The delay between the clock pulse going high and the valid output of the first gate will cause the second gates, output to glitch or show an invalid logic output. If we can hold the output voltage of the PE gate low instead of high, we can eliminate this race condition. Upon adding an inverter to the PE gate (Fig. 14.15), the condition for glitch-free operation is met. The PE gate with the addition of an inverter is called Domino logic. The name *Domino* comes from the fact that a gate in a series of Domino logic gates cannot change output states until the previous gate changes states. The change in output of the gates occurs similar to a series of falling dominoes. The inverter used in the Domino gate has the added advantage that it can be sized to drive large capacitive loads.

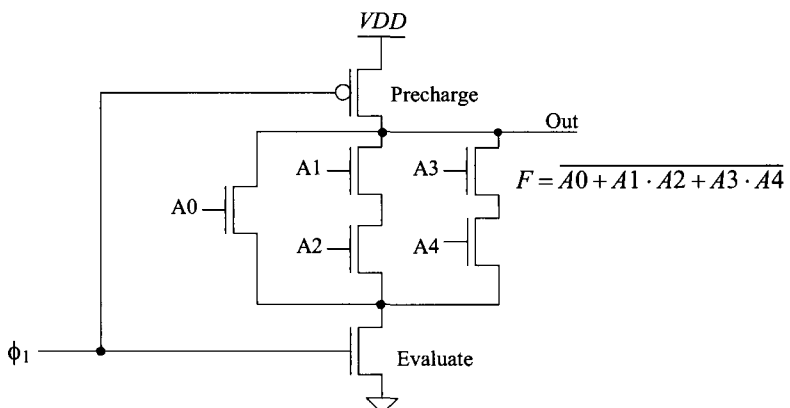


Figure 14.13 A complex PE gate.